

The exact support barrier for universal relabeling averages

An application of Klyachko's classical constituent theorem

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Let $L_{n,k}$ be the degree- k part of the reduced free Lie algebra on n labeled generators, where every bracket with a repeated letter vanishes. The symmetric group S_n acts by relabeling generators. For a probability measure μ on S_n , put

$$T_\mu = \sum_{g \in S_n} \mu(g)g.$$

Call μ a universal relabeling annihilator if T_μ acts as zero on $L_{n,k}$ for every $2 \leq k \leq n$.

Theorem 1. *For $n \geq 4$, the universal relabeling annihilators are exactly*

$$\mu(g) = \frac{1 + c \operatorname{sgn}(g)}{n!} \quad (-1 \leq c \leq 1).$$

Their minimum support size is $n!/2$, attained only by the uniform measure on the even permutations or on the odd permutations. For $n = 2, 3$, the unique annihilator is the uniform measure on S_n . For $n \geq 7$, and also for $n = 5$, annihilating just the top degree $L_{n,n}$ already forces the displayed formula.

Lemma 2 (Constituents). *For $n \geq 4$, the union of the irreducible constituents of the modules $L_{n,k}$, $2 \leq k \leq n$, consists of every irreducible S_n module except the trivial and sign representations. For $n = 2, 3$, it consists of all nontrivial irreducibles.*

Proof. Write Lie_k for the multilinear free Lie representation of S_k . Decomposition by the set of letters occurring in a bracket gives

$$L_{n,k} \cong \operatorname{Ind}_{S_k \times S_{n-k}}^{S_n} (\operatorname{Lie}_k \boxtimes \mathbf{1}).$$

Klyachko's constituent theorem says that for $k > 1$, Lie_k contains all Specht modules of size k except the trivial module, the sign module when $k > 2$, and the two exceptional shapes $(2, 2)$ and $(2, 2, 2)$. See [1, Theorem 1.1, Theorem 1.2 and Remark 1.3] for the exact classification and its connection to the multilinear Lie representation, and [2] for a combinatorial proof of the Lie constituent theorem. Thus for $n \geq 7$ or $n = 5$, the top degree alone contains every irreducible except trivial and sign. For $n = 4$, the missing shape $(2, 2)$ occurs in $L_{4,3} = \operatorname{Ind}_{S_3}^{S_4} \operatorname{Lie}_3$, since $\operatorname{Lie}_3 = S^{(2,1)}$ and one-box branching adds $(2, 2)$. For $n = 6$, the missing shape $(2, 2, 2)$ occurs in $L_{6,5}$ since $S^{(2,2,1)}$ occurs in Lie_5 and one-box branching applies.

No $L_{n,k}$ contains the trivial representation: Frobenius reciprocity reduces its multiplicity to that of the trivial representation in Lie_k , which is zero for $k \geq 2$. For $n \geq 4$, none contains sign either. If $n - k \geq 2$, its restriction has nontrivial sign on the S_{n-k} factor, incompatible with $\mathbf{1}$ there. If $n - k \leq 1$, then $k \geq 3$, and Lie_k itself has no sign constituent. This proves the claim for $n \geq 4$. For $n = 2$, Lie_2 is sign. For $n = 3$, $L_{3,3}$ is the standard module, while $L_{3,2} = \operatorname{Ind}_{S_2}^{S_3} \operatorname{sgn}$ contains both standard and sign. $\square$

Proof of Theorem 1. Work over $\mathbb{C}$. If an irreducible module occurs in a module on which T_μ vanishes, the matrix Fourier transform $\sum_g \mu(g)\rho(g)$ is zero on that irreducible. Conversely, vanishing on every constituent implies vanishing on the whole module. Lemma 2 therefore says that for $n \geq 4$, all Fourier components of μ vanish except possibly those belonging to trivial and sign. Fourier inversion on the finite group yields

$$\mu(g) = \frac{1 + c \operatorname{sgn}(g)}{n!}, \quad c = \sum_g \mu(g) \operatorname{sgn}(g).$$

Normalization supplies the coefficient one of the trivial component; nonnegativity is equivalent to $-1 \leq c \leq 1$. For $|c| < 1$ all $n!$ values are positive; for $c = \pm 1$ exactly one parity class is supported. Every such measure annihilates the listed constituents, proving the converse and the support assertion. The small cases and top-degree assertion follow from the same constituent calculation. $\square$

Consequences and limits. There is no universal positive relabeling average with support $\exp(O(n))$. In fact its support is at least $n!/2$, even if only the highest-degree error must be annihilated when $n \geq 7$. This is a barrier for an operator required to annihilate *every* possible error in the stated Lie modules. It is not a lower bound for an average tailored to a single error vector, for nonlinear operations, or for the size of permutation-fair dice.

Uniform averaging over A_n can replace averaging over all S_n in the universal fair-prefix completion construction when $n \geq 4$. The identity permutation is present, so the literal prefix is preserved. This halves each symmetrization multiplier but does not change its factorial asymptotic scale. Relabeling generators is the group action used throughout; rotation of positions inside a word is a different operation and is not covered by this theorem.

Exact finite checks. The script `check_lie_constituents.py` independently counts standard tableaux by major index using corner recursion. For every $2 \leq n \leq 20$, it verifies the stated exceptional shapes and the identity $\sum_\lambda d_\lambda m_\lambda = (n-1)!$. These computations corroborate the cited theorem; the proof above does not depend on a finite-range extrapolation.

Attribution. The constituent theorem and finite-group Fourier decomposition are classical. The statement above is their direct application to the universal averaging mechanism in the dice problem. Historical novelty of this precise application has not been established.

References

- [1] Joshua P. Swanson, *On the existence of tableaux with given modular major index*, Algebraic Combinatorics 1 (2018), 3–21, doi:10.5802/alco.4.
- [2] L. G. Kovács and Ralph Stöhr, *A combinatorial proof of Klyachko’s theorem on Lie representations*, Journal of Algebraic Combinatorics 23 (2006), 225–230, doi:10.1007/s10801-006-7394-6; author’s full text.